# KLS OneMill — Milling Module User Guide

This is the user documentation for the **Milling** module of KLS OneMill — the part of the application that turns a positioned dental implant on its raw part into a validated, machine-ready NC program.

**Scope: KLS OneMill 2.1.0.** This guide intentionally covers only what has actually shipped — features still in progress on a feature branch (currently anything from the 2.2.0 release notes, e.g. multiple implants per raw part, raw part marking, VHF Titan milling) are left out until they release, rather than documented as "(Preview)". See "Keeping this document alive" below for the update process each release.

It is a **living document**: it describes the workflow and features as they exist in the current release, and is meant to be updated alongside the module rather than rewritten from scratch. Screenshots are not included yet — every place one would help is marked with a placeholder like the one below, ready to be filled in:

> 🖼️ *Screenshot: placeholder — add a capture of the relevant panel/dialog here.*

## Contents


| Page | Covers |
|---|---|
| 01 — Overview & Workflow | What the module does, the session state machine, the workflow toolbar |
| 02 — Input Settings | Region/Material/Machine/Fixture, strategy selection |
| 03 — Raw Part & Implant | Raw part setup, the Raw Part Library, placement tools |
| 04 — Holes, Csys & Milling Templates | Hole detection, per-hole machining strategy, angle/position validation, IGES & Open Milling templates |
| 05 — Struts & Links | Manual struts, per-strut shape and milling-side settings |
| 06 — Strategy Tree & Session | The session tree, dynamic values, resetting entries/strategies |
| 07 — Generation, Validation & Toolpath | The Generate step, pre-generation checks, the Operations tree |
| 08 — Output & Post-Processing | The Post step, what leaves the module, VHF-specific behavior |
| 09 — Configuration Reference | Admin-level settings: the Milling config tab, machines, fixtures, templates |
| 10 — Extended Settings | Bulk strut defaults, milling-window/surface offsets, Special Options |


## Who this is for

Written for the person actually running a milling session day to day — planning an implant, placing it on a raw part, choosing how each hole and strut gets machined, and generating and posting the result. Pages 01–08 and 10 describe that workflow. Page 09 is aimed more at whoever maintains machine/fixture/template configuration for the shop.

## Keeping this document alive

- **On each new release**, walk the "Milling" section of the matching release notes top to bottom and, for each entry that changes what a user sees or does, update the page that already covers it (add a new page only once an existing one has grown unwieldy). A feature still marked "(Preview)" in the release notes for the release you're documenting is still fair game to include — this guide tracks what's shipped, not what's finished changing shape.
- Add a screenshot wherever you see the 🖼️ placeholder — a capture of the actual panel/dialog is enough, no annotation required. Run `build_pdf.py` (see below) after updating placeholders to refresh the exported PDF.
- Bumping the documented scope to a new release is a content change, not a restructuring — resist renumbering pages unless one is actually added or removed.

## Generating the PDF

`build_pdf.py` (Python 3, needs the `markdown` package: `py -m pip install markdown`) concatenates this README and pages 01–10 in order and prints the result to `KLSOneMill-Milling-UserGuide.pdf` using Chrome's built-in headless PDF printer — no other dependency (pandoc, LaTeX, …) required. From this folder:

```
py build_pdf.py
```

Pass `--chrome "C:\path\to\chrome.exe"` if Chrome isn't found automatically, or `--out some-name.pdf` to write elsewhere. Add a new page to the `PAGES` list at the top of the script when the contents table above gains one.

# 01 — Overview & Workflow

## What the Milling module does

The Milling module ("Simple Milling") is where a dental implant, positioned on its raw part, goes from a positioned model to a validated, machine-ready NC program. It owns the raw part, the implant's placement on it, hole and strut detection, per-hole/per-strut machining strategy, geometry and toolpath generation, and hands the result to the Post Processor module for final NC output.

> 🖼️ *Screenshot: the Milling module's main layout — 3D viewport, Input Settings panel, session tree, workflow toolbar.*

## The session workflow bar

Across the top of the module sits a row of buttons that double as a state indicator — a colored dot (red while idle, green once a workpiece is loaded) shows whether the session has data to work with. The buttons are meant to be used roughly left to right, though later ones only enable once the earlier step has actually produced something to build on:

1. **AssignImplantPos** — commits the implant's current position (from the interactive transform gizmo) as its working placement. Available as soon as an implant is loaded.
2. **Refresh** — re-runs workpiece position validation (hole/csys angle checks, colors) without regenerating geometry. Useful after nudging the implant or switching machine/fixture.
3. **Geometry** — builds the milling geometry for the current implant: contours, milling windows, detected holes, struts. This is the step that turns "an implant sitting on a raw part" into something the Strategy tree and hole/strut tools can actually act on.
4. **Generation** — runs toolpath generation for the current implant, after a confirmation dialog ("Bone Surface Side checked? Correct strategy?") that exists specifically to catch the most common setup mistake before committing to a run.
5. **Result** — switches to the generated toolpaths for review (the session tree's Operations view) once Generation has produced them.
6. **Post** — hands the session to the Post Processor module to produce the actual NC file(s). See 08 — Output & Post-Processing.

> 🖼️ *Screenshot: the workflow toolbar in a mid-session state (Geometry done, Generation enabled).*

Each button's tooltip explains what's still missing when it's disabled (e.g. "Load implant first", "Generate geometry first"), so the toolbar itself is a reasonable first place to check when something further along seems unavailable.

## Where things live on screen

- **Input Settings** (top of the panel area) — region, material, machine, fixture, raw part, and per-implant workpiece settings. See 02 — Input Settings.
- **3D Viewport** — the raw part, implant(s), holes/csys markers, struts, milling windows, and (once generated) the toolpath itself. Holes, csys markers, and struts are directly clickable to open their own per-item settings.
- **Session tree** (bottom toolbox area, expands when the module is active) — a structured view of the whole session: setup, geometry, raw part, the assigned machining strategy and its entries, and the generated Operations. See 06 — Strategy Tree & Session.
- **Graphics toolbar overlay** — floats above the 3D view and picks up controls specific to what's currently displayed. See 07 — Generation, Validation & Toolpath.

## A typical session, end to end

1. Pick Region / Material / Machine / Fixture and load the implant ( 02 — Input Settings).
2. Set up the raw part — dimensions and material ( 03 — Raw Part & Implant).
3. Click **AssignImplantPos**, then **Geometry** to detect holes and build the milling geometry.
4. Review detected holes/csys markers, assign machining strategies where the default doesn't fit, and add any manual struts (04, 05).
5. Adjust dynamic values or optional entries on the assigned strategy if needed ( 06 — Strategy Tree & Session).
6. Click **Generation**, resolve anything the pre-generation validation flags, and review the result ( 07 — Generation, Validation & Toolpath).
7. Click **Post** to produce the NC output for the target machine ( 08 — Output & Post-Processing).

# 02 — Input Settings

The Input Settings panel is where a session's fixed context — region, material, machine, fixture — and the raw part / implant data live. It sits above the 3D viewport.

> 🖼️ *Screenshot: Input Settings panel.*

## Session-wide fields

**Region**, **Material**, **Machine**, and **Fixture** are chosen from dropdowns that filter each other — the material list reflects what the selected region supports, the machine list reflects region+material, and so on. **Titan**, **Peek**, and **MultiPurpose** are the available material types.

Machines (and, by extension, any region/material combination only reachable through one) flagged as test-only in their configuration are hidden from these lists during normal use — they only appear once the application is switched into Test mode. The same applies to machining strategies flagged as test-only. This keeps in-development machine setups out of the way without deleting them.

Switching the machine re-validates the current workpiece position (hole/csys angle checks against the new fixture) immediately, rather than waiting for the next **Refresh**.

## Machining strategy selection

Each implant has its own assigned machining strategy (a factory-standard template for the current machine/material). Changing it updates the session's strategy tree — see 06 — Strategy Tree & Session for what that tree shows and how to adjust individual entries afterward.

## Raw part

The raw part group covers plate dimensions (thickness/width/height) and material — see 03 — Raw Part & Implant.

> 🖼️ *Screenshot: Raw Part group, showing dimensions.*

## Work-piece (implant)

The work-piece group covers loading the implant's file and shows the detected base shape once loaded.

## Extended Settings

A separate **Extended Settings** panel groups the less-frequently-touched controls: bulk strut defaults, milling-window/surface offset fields, and the action buttons **AddManual Strut**, **Generate EdgePlanes**, **Detect Holes**, and **Remove Detected Holes**. See 10 — Extended Settings for the full breakdown.

> 🖼️ *Screenshot: Extended Settings panel.*

# 03 — Raw Part & Implant

## The raw part

A raw part is the physical plate that gets loaded into the machine. Its dimensions (thickness/width/height) and material are set in the Raw Part group of Input Settings — see 02 — Input Settings.

## Placing the implant

Dragging the implant shows a live outline of its footprint on the raw part's mounting plane. This preview updates continuously while dragging, not just once you let go.

**Suggest Position** looks for a free spot for the implant automatically, respecting the configured inter-implant clearance and whatever area a reused plate already reports as occupied (see below).

## Raw Part Library

Instead of starting a fresh plate every time, a raw part can be **saved to** and **loaded from** a library of named plates:

- **Save to Library** stores the plate's machine/fixture/dimensions and a snapshot of where the implant currently sits, together with a thumbnail.
- **Load from Library** opens a browser of previously saved plates (dimensions and thumbnail shown per entry) to reuse.
- **Reuse Raw Part** is the lighter-weight alternative: an ad-hoc folder picker over any earlier session's own working folder, with no catalog or thumbnail — useful for quickly picking up where a specific past session left off rather than a curated, named plate.

> 🖼️ *Screenshot: Raw Part Library browser dialog.*

Reusing a plate (either way) correctly restores its actual dimensions and whatever area was already milled/occupied on it — placement tools and **Suggest Position** treat that occupied area as unavailable.

## Materials

**Titan**, **Peek**, and **MultiPurpose** are the available raw part material types, selected through the same region/material filters used everywhere else in the session.

# 04 — Holes, Csys & Milling Templates

## Detection

**Detect Holes** (Extended Settings) finds drilling holes and coordinate-system markers (csys) on the loaded implant automatically. **Remove Detected Holes** clears them if you need to redo detection after a change. Detection parameters are admin-level settings, see 09 — Configuration Reference.

## Assigning how a hole or csys gets machined

Clicking a hole or csys marker in the 3D viewport opens a persistent panel (not a popup menu) with a combo box and checkboxes for that item. From here you can:

- Assign a specific machining strategy ("hole norm") to just this hole/csys, overriding whatever the machine's default would otherwise use.
- Enable or configure an **Open Milling Template** override for it — a cone-shaped pocket shape, configured per machine as a named, reusable set (see 09 — Configuration Reference).

Confirm with OK to commit, or Cancel to discard. Committing an override updates the on-screen preview/coloring immediately, but does **not** by itself rebuild the full milling geometry — click **Geometry** (or, if geometry was already generated, the change may regenerate automatically for that one geometry element) to see the actual effect in the toolpath.

> 🖼️ *Screenshot: hole/csys settings panel with the machining-strategy combo and Open Milling Template checkbox.*

The machine-wide default strategy for holes/csys can also be changed from Input Settings — that changes the fallback used by every hole/csys that doesn't have its own explicit override.

## Angle and position validation

Before generating, and whenever **Refresh** runs, each hole's axis and each csys's normal direction is checked against the current fixture's configured angle tolerance (currently a single tolerance, the same for every direction). A hole/csys that fails is drawn in the error color (applied to every visible part of its marker, not just the outline) so it's obvious at a glance which ones need attention. Switching machine/fixture re-runs this check immediately.

Separately, a **reference-point check** can verify that each hole's/csys's reference point sits on the correct side of the implant surface along its normal (near or far side, set per machining entry) before generation starts — this stays reliable even on imperfect geometry, since it raycasts against the real surface rather than requiring an exact positional match. Anything that fails either check is listed in a **Validation Warning** dialog before

generation, with the option to continue anyway. A separate, off-by-default setting can also show a confirmation dialog when validation finds *no* issues at all, giving a chance to double-check (or cancel) even a clean run.

> 🖼️ *Screenshot: Validation Warning dialog listing failed holes/csys.*

## IGES-driven milling passes

A machining strategy entry can drive its geometry from an authored IGES file instead of (or in addition to) built-in shapes:

- A single **curve** or **surface** reference for straightforward wireframe/surface milling.
- Multiple typed inputs at once — a trimmed surface plus detected top/center/bottom edge curves from that same surface's geometry. When both a top and bottom edge curve are present, the pass mills by morphing between the two curves rather than following the full surface or a single wireframe.

# 05 — Struts & Links

Struts (also called links) connect the implant to the raw part or bridge between milling features so the part stays attached — and machinable — through the whole job.

## Placing struts manually

**AddManual Strut** (Extended Settings) arms a placement tool — click on the implant/raw part to drop a strut there. A newly placed strut starts out using the session's current bulk strut defaults (count, shape, orientation, sizing) — see 10 — Extended Settings — before any of its own per-strut overrides below are applied.

> 🖼️ *Screenshot: manual strut placement.*

## Editing a strut

Each strut has its own context menu (right-click, or the per-strut settings panel) covering:

- **Shape** — round or rectangular cross-section, with the matching dimension fields (diameter, or width/height).
- **Orientation** and **shift value** along the contour.
- **Mill bone side** / **Mill opposite side** — independent toggles for whether each side of the strut gets machined.
- **Offset** — how far the strut's milling extends outward.

A strut can also be repositioned or rotated directly with the interactive transform gizmo in the 3D view; editing its shape afterward keeps that manual placement rather than resetting it.

> 🖼️ *Screenshot: strut context menu with shape/orientation/mill-side fields.*

# 06 — Strategy Tree & Session

## The session tree

The bottom toolbox area holds a tree view of the entire session, expanding automatically while the Milling module is active:

1. **Milling Setup** — machine and fixture.
2. **Geometry** — implant, detected holes, curves, csys.
3. **Raw Part**.
4. **Strategy** — the assigned machining strategy, its entries, and each entry's dynamic values.
5. **Operations** — the generated toolpaths, once Generation has run (see 07 — Generation, Validation & Toolpath).

> 🖼️ *Screenshot: session tree showing Strategy entries expanded with dynamic values.*

## Editing values from the tree

Selecting a dynamic-value item installs an inline editor for it — a spin box for a numeric value, a checkbox for a boolean one — right in the tree, with its range (if any) taken from that value's configured min/max. Changes apply immediately, no separate "commit" step.

Some strategy entries are **optional**: they show their own checkbox to switch that entry on or off entirely for this session, independent of every other entry.

## Resetting values

Right-clicking gives context-appropriate reset actions:

- On a dynamic value that's been changed from its default: **Reset Value**.
- On a strategy entry: **Reset Entry** — resets every dynamic value inside that one entry.
- On the strategy itself: **Reset Strategy** — resets everything under it back to the template's defaults.

Other tree items (geometry, raw part, etc.) show separate **Show** and **Hide** context-menu entries reflecting their current visibility — only the applicable one of the pair is enabled at a time. Individual generated-operation entries under Operations additionally offer **Show Toolpath** — see 07 — Generation, Validation & Toolpath for what that displays.

## Per-strategy post-processor variables

A machining strategy can carry its own set of post-processor variables — for example, a machine option that should only take effect for sessions using that one specific strategy. These are merged with the machine's regular post-processor settings when the NC program is generated, without changing anything about the machine's own configuration for other sessions.

# 07 — Generation, Validation & Toolpath

## Starting generation

Clicking **Generation** first asks a single deliberate confirmation — "Bone Surface Side checked? Correct strategy?" — before doing anything. It exists to catch the single most common setup mistake before committing to a run that can take a while and produce a lot of downstream output. Answering **Cancel** aborts with nothing generated.

Before a new generation actually starts, the application also checks whether the selected working directory already contains files from an earlier run, and offers the choice to overwrite the existing output or write into a new, timestamped directory instead — protecting against accidentally clobbering a previous result.

> 🖼️ *Screenshot: the "Bone-Surface-Correct?" confirmation dialog.* 🖼️ *Screenshot: working-directory overwrite-or-new-folder dialog.*

## Pre-generation validation

Before toolpaths are actually built, the session is checked for:

- **Hole/csys angle**, against the active fixture's configured tolerance.
- **Hole/csys reference-point position**, checking each one sits on the correct side of the implant surface for its configured direction.
- Anything else flagged as required by the assigned machining strategy.

Anything that fails is listed in a **Validation Warning** dialog, with the option to continue anyway rather than being hard-blocked. A separate, off-by-default setting (Milling config tab → Settings) can additionally show a confirmation dialog when validation finds **no** issues, giving a deliberate chance to still cancel even a fully clean run.

> 🖼️ *Screenshot: Validation Warning dialog.*

## Reviewing the result

Clicking **Result** switches the session tree to its **Operations** view — a list of the generated toolpath operations for the current implant. Selecting **Show Toolpath** on an operation displays it in the 3D viewport.

> 🖼️ *Screenshot: Operations tree after generation.*

# 08 — Output & Post-Processing

## Handing off to the Post Processor

Clicking **Post** activates the Post Processor module for the current implant/session, which is what actually writes the machine-readable NC program from the toolpaths Generation produced.

> 🖼️ *Screenshot: Post Processor module after clicking Post from a generated Milling session.*

The Post Processor module is a separate plugin with its own settings and output options (job file selection, machine-specific variables, output directory); this page only covers what's relevant from the Milling side of that handoff. Detailed Post Processor usage belongs in that module's own documentation once written.

## Machine-specific behavior worth knowing about

- **VHF machines** carry configurable output options for water processing, ionizer, preset raw-part number, estimated milling time, and material label, plus a configurable output file name (instead of a fixed `main.nc`). The estimated milling time is calculated once generation completes and written into the NC header automatically.
- A **toolbreak check** is emitted before a tool-change block whenever the preceding operation was a drilling operation, and at end-of-program when the last operation was a drilling operation.
- The generated post-processor job data includes the current Windows user name, available to post-processor scripts (e.g. for NC header information).

## Advanced/administrative note

Authoring the machine/fixture/post-processor configuration itself is shop/administrator-level work rather than part of the everyday per-implant workflow — see 09 — Configuration Reference.

# 09 — Configuration Reference

This page is aimed at whoever maintains machine, fixture, and template configuration for the shop, rather than day-to-day session work. It's deliberately lighter on step-by-step instructions and heavier on "what exists and where to find it," since this configuration changes far less often than a normal session.

## Milling config tab

The application's configuration dialog has a dedicated **Milling** tab grouping module-specific options in one place:

- Post-processor output settings.
- Hole-detection parameters (used by **Detect Holes**, see 04 — Holes, Csys & Milling Templates).
- **Settings** → "Show info message when validation is successful" — off by default; shows a confirmation dialog before generation whenever pre-generation validation finds no issues at all (see 07 — Generation, Validation & Toolpath).

> 🖼️ *Screenshot: Milling tab in the application configuration dialog.*

## Test mode

Machines, and machining strategies, can each be individually flagged as test-only. Flagged entries — along with any region/material combination only reachable through one — are hidden from the normal selection lists and only appear once the application is switched into **Test** mode from the configuration window. Test mode is not persisted between sessions; it always resets back to normal on the next application start.

## Fixtures

Each fixture defines its own hole/csys angle-check tolerance (see 04 — Holes, Csys & Milling Templates).

## Machining strategies & templates

- **Factory-standard templates** define the default machining approach per machine/material and can reference IGES-authored geometry or pre-posted subprogram files as their source, instead of only built-in shapes.
- **Open Milling Templates** (the cone-shaped pocket shapes referenced from 04) are configured per machine as a named, reusable set.

# 10 — Extended Settings

The **Extended Settings** panel holds the controls that don't need to be visible all the time: bulk strut defaults, the milling-window/surface offsets that shape the generated geometry, and a small set of administrator tools.

> 🖼️ *Screenshot: Extended Settings panel.*

## Action buttons

- **AddManual Strut** — arms the strut-placement tool; click a point on the implant/raw part to drop a strut there. See 05 — Struts & Links for placing and editing individual struts.
- **Generate EdgePlanes** — builds the chamfer/edge-milling planes for edge-plane milling strategy entries, from the raw part's geometry and the configured angle/offset.
- **Detect Holes** *(currently labeled "Detect Holes (Peek-Beta)" in the UI, reflecting its current material scope)* — runs automatic hole/csys detection on the current implant.
- **Remove Detected Holes** — clears detection results so it can be re-run after a change.

> 🖼️ *Screenshot: the action buttons in Extended Settings.*

## Strut Settings

This group holds the **bulk defaults** applied to every newly created strut and to a "change all struts at once" edit:

- **Strut Count** — how many struts a bulk (non-manual) strut generation places.
- **Strut Start-Offset** — where along the contour the first strut is placed.
- **Strut Direction Offset** — shift value along the contour applied to every strut.
- **Strut Geometry Inside** — how far a strut is moved inward from the contour.
- **Strut Orientation** — **MeshNormal** or **MillingPlane**, i.e. whether a strut's own orientation follows the implant's mesh surface or the flat milling plane.
- **Strut Shape** — **Round** or **Rectangle**; picking one reveals the matching size fields below it (**Strut Size Inner/Outer** for Round, **Strut Size Inner/Outer (w×h)** for Rectangle).
- **NextGenStruts** — selects the newer, more robust strut-generation engine for this session instead of the legacy one. Leave this on unless you have a specific reason to fall back to the legacy behavior.
- **GeometryCenter** — centers a strut's cross-section on the contour surface rather than the older placement method.

> 🖼️ *Screenshot: Strut Settings, Round shape fields visible.*

These are **global, per-session** defaults — they're what a freshly placed or bulk-regenerated strut uses. An individual strut's own context menu (see 05 — Struts & Links) can still override shape, orientation, or offset for just that one strut without touching these session-wide values.

## Window / Surface Offset Settings

This group holds the offsets that shape the generated milling window and its associated surfaces:

- **ContourSimplification Offset** — how aggressively the detected contour is simplified before building the milling window from it.
- **Main Window Offset** — how far the main milling window extends outward from the implant contour.
- **Center Surface Inner Offset** — inward offset for the center working surface.
- **Depth Surface Direction Offset** *(shown as "Dept Surface Direction Offset" in the current UI)* — offset along the milling direction for the depth/center surface.
- **BottomSurface Geometry Offset** — offset for the bottom working surface. Only visible in Test mode; hidden otherwise.

> 🖼️ *Screenshot: Window / Surface Offset Settings.*

## Special Options

A group labeled **"Special Options"** holds two administrator tools plus one behavior switch:

- **ParseParameter** — picks an input and an output directory and converts legacy parameter files between them. A one-off migration tool, not part of everyday session work.
- **Default Configuration** — picks an output directory and writes a fresh set of default configuration files into it. Also an administrator tool — see 09 — Configuration Reference for the configuration this feeds into.
- **Whether this group is expanded also changes what AddManual Strut does**: with Special Options expanded, a manually placed strut attaches to the currently selected contour curve; with it collapsed (the default), a manually placed strut is placed independently of any curve. This isn't labeled as a strut option in the UI today — if a manually placed strut isn't behaving the way you expect, check whether this group happens to be open.

> 🖼️ *Screenshot: Special Options group expanded.*